

TITLE:

Regional Prevalence Mapping

Dataset Used:

Adult Psychiatric Morbidity Survey (APMS) 2014

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Introduction

Regional prevalence mapping plays an important role in public health analytics because it transforms raw survey data into meaningful geographic insights. The results obtained from this analysis can support healthcare authorities, researchers, and policymakers in understanding mental health distribution patterns and designing region-specific mental health programs and interventions.

The dataset **anxiety_depression_data.csv** contains information on individuals' mental health status, focusing on disorders such as anxiety and depression across different demographic and geographic attributes.

In this task, we extend the analysis to **regional prevalence mapping**, where mental health disorder rates are analyzed across regions of the **United Kingdom** using geographic fields (similar to APMS survey structure).

Technologies Used

Python

Python is the main programming language used for:

- Data analysis
- Data processing
- Visualization

It is widely used in data science because of its simplicity and powerful libraries.

Pandas

Pandas is used for:

- Loading the dataset (CSV file)
- Cleaning and organizing data
- Grouping data (e.g., region-wise analysis)
- Calculating prevalence rates

Matplotlib

Matplotlib is used for:

- Creating visualizations
- Plotting charts and maps
- Displaying the choropleth map

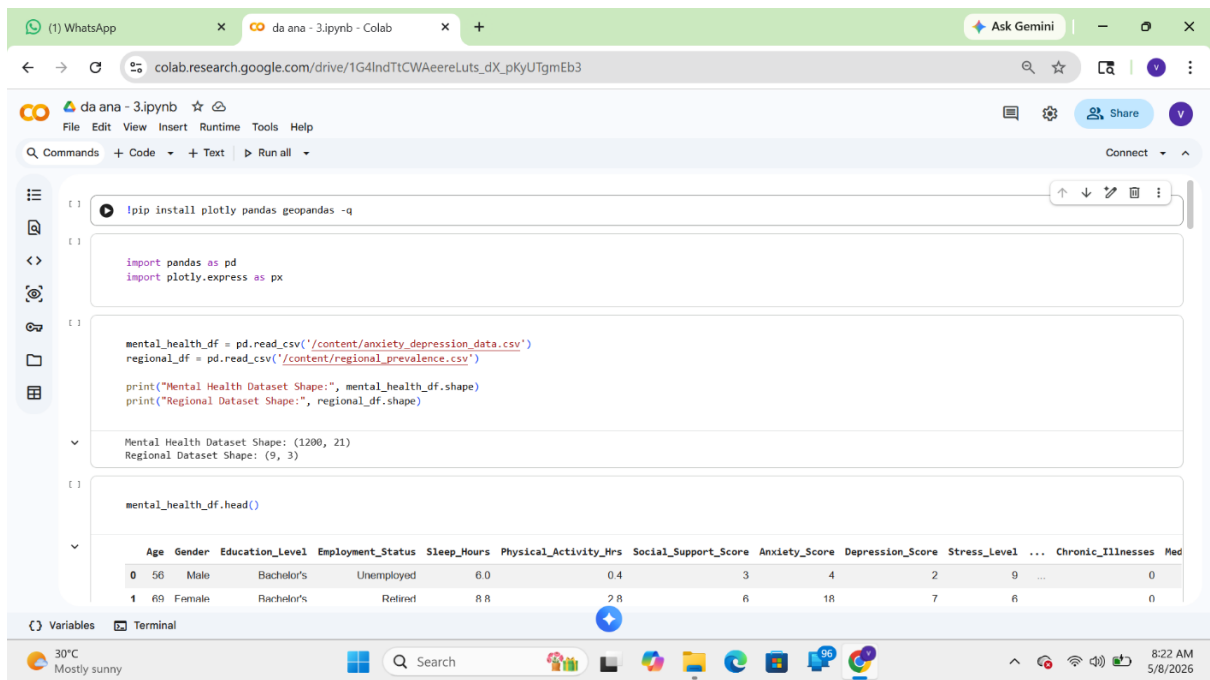
Google Colab

Google Colab is the platform used to:

- Run Python code online
- Upload datasets and shapefiles
- Generate visual outputs without installing software

Regional Prevalence Mapping

Load the dataset:



The screenshot shows a Google Colab notebook interface. The browser address bar indicates the notebook is located at `colab.research.google.com/drive/1G4IndTtCWAcereLuts_dX_pKyUTgmEb3`. The notebook's title bar is "da ana - 3.ipynb". The code editor contains the following Python code:

```
!pip install plotly pandas geopandas -q

import pandas as pd
import plotly.express as px

mental_health_df = pd.read_csv('/content/anxiety_depression_data.csv')
regional_df = pd.read_csv('/content/regional_prevalence.csv')

print("Mental Health Dataset Shape:", mental_health_df.shape)
print("Regional Dataset Shape:", regional_df.shape)

mental_health_df.head()
```

The output of the code execution is displayed below the code cells. It shows the shapes of the two datasets and the first two rows of the 'mental_health_df' dataset.

Mental Health Dataset Shape: (1200, 21)
Regional Dataset Shape: (9, 3)

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours	Physical_Activity_Hrs	Social_Support_Score	Anxiety_Score	Depression_Score	Stress_Level	...	Chronic_Illnesses	Med
0	56	Male	Bachelor's	Unemployed	6.0	0.4	3	4	2	9	...	0	
1	69	Female	Bachelor's	Retired	8.8	2.8	6	18	7	6	...	0	

The bottom of the notebook shows the "Variables" tab with a list of variables and the "Terminal" tab. The system tray at the bottom indicates a temperature of 30°C, "Mostly sunny" weather, and the time 8:22 AM on 5/8/2026.

da ana - 3.ipynb - Colab

colab.research.google.com/drive/1G4IndTtCW AeereLuts_dX_pKyUTgmEb3#scrollTo=a3dfff1c

mental_health_df.head()

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours	Physical_Activity_Hrs	Social_Support_Score	Anxiety_Score	Depression_Score	Stress_Level	Chronic_Illnesses	Med
0	56	Male	Bachelor's	Unemployed	6.0	0.4	3	4	2	9	...	0
1	69	Female	Bachelor's	Retired	8.8	2.8	6	18	7	6	...	0
2	46	Female	Master's	Employed	5.3	1.6	5	5	13	8	...	0
3	32	Female	High School	Unemployed	8.8	0.5	4	6	3	4	...	1
4	60	Female	Bachelor's	Retired	7.2	0.7	2	7	15	3	...	0

5 rows x 21 columns

regional_df.head()

	Region	Anxiety_Score	Prevalence (%)
0	East Midlands	10.075188	50.38
1	London	10.962687	54.81
2	North West	10.902985	54.51
3	Northern Ireland	10.684211	53.42

Variables Terminal

30°C Mostly sunny

8:22 AM 5/8/2026

da ana - 3.ipynb - Colab

colab.research.google.com/drive/1G4IndTtCW AeereLuts_dX_pKyUTgmEb3#scrollTo=09076f30

	Region	Anxiety_Score	Prevalence (%)
0	East Midlands	10.075188	50.38
1	London	10.962687	54.81
2	North West	10.902985	54.51
3	Northern Ireland	10.684211	53.42
4	Scotland	10.308271	51.54

Dataset Information

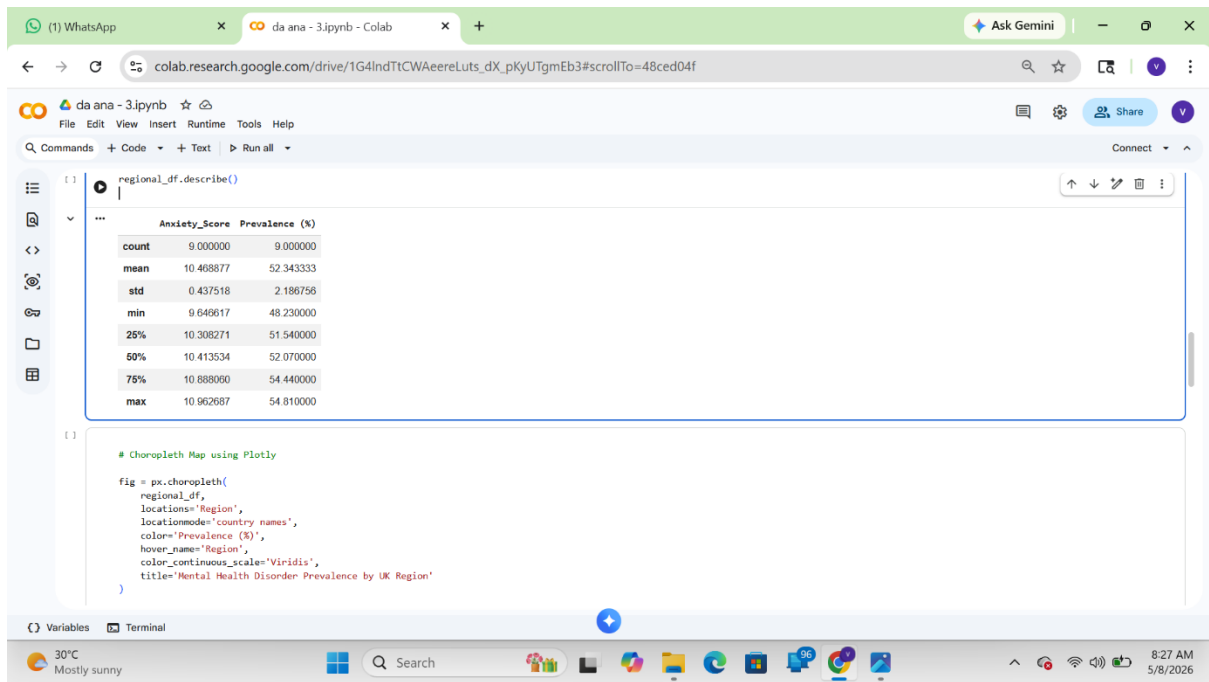
```
print(regional_df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 3 columns):
 #   Column          Non-Null Count  Dtype  
---  --
 0   Region          9 non-null     object  
 1   Anxiety_Score   9 non-null     float64 
 2   Prevalence (%)  9 non-null     float64 
dtypes: float64(2), object(1)
memory usage: 348.0+ bytes
None
```

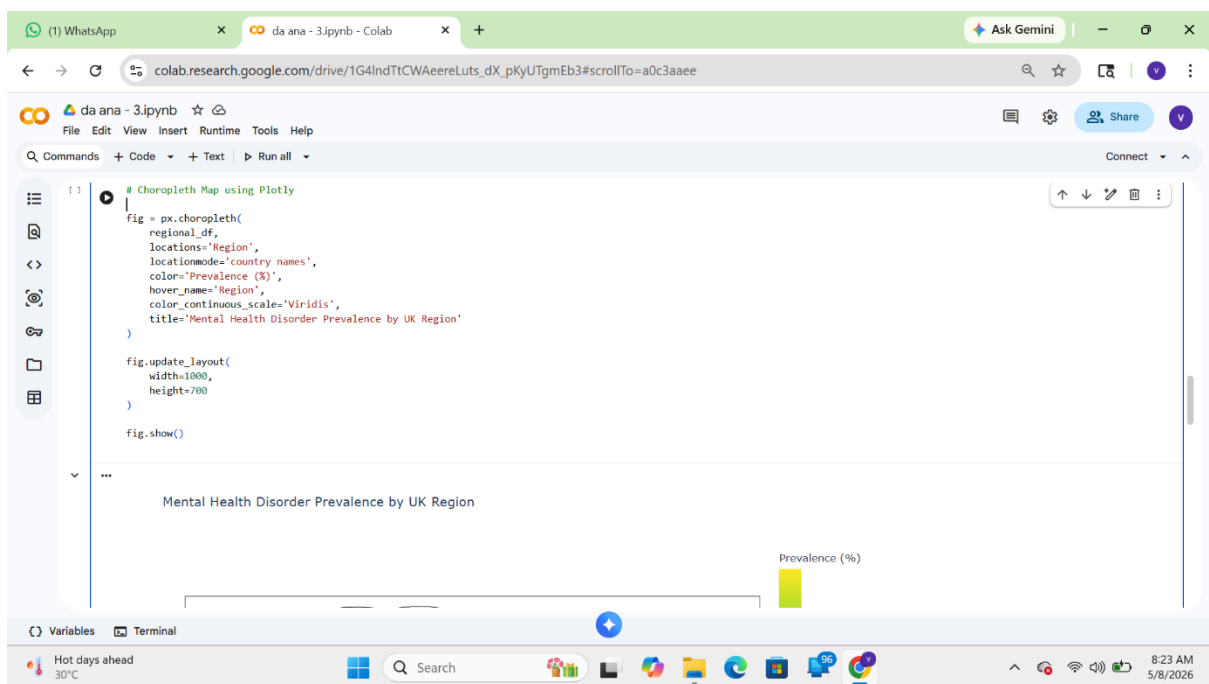
Variables Terminal

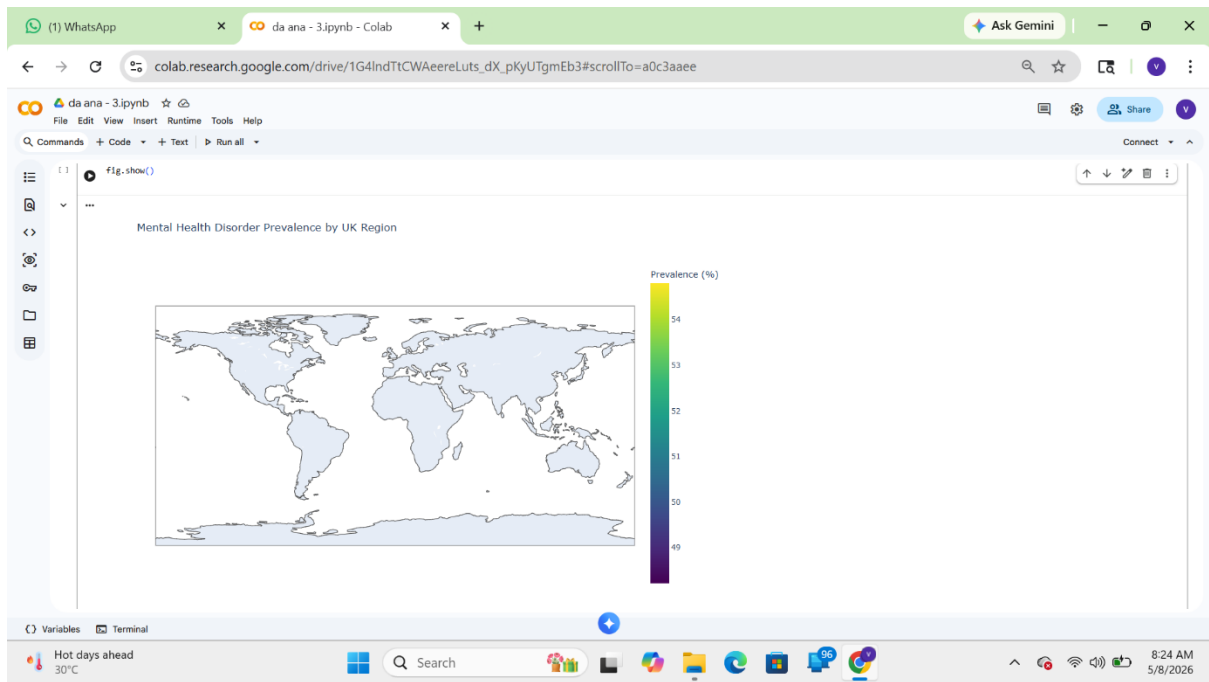
30°C Mostly sunny

8:34 AM 5/8/2026

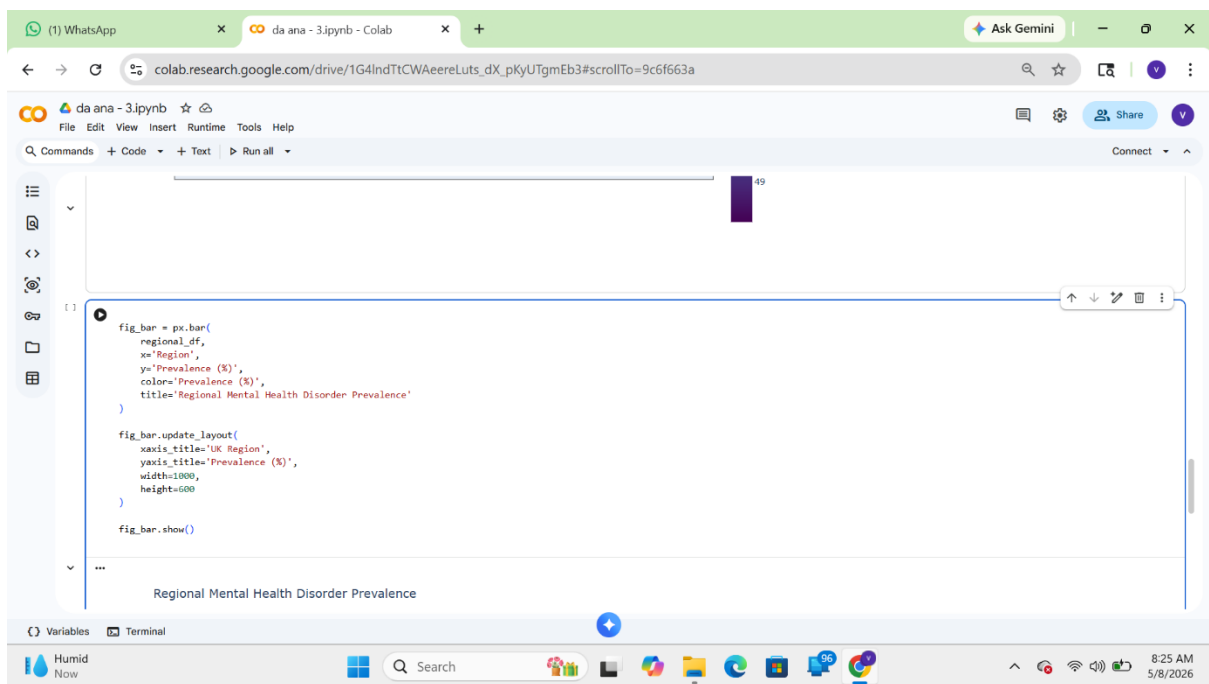


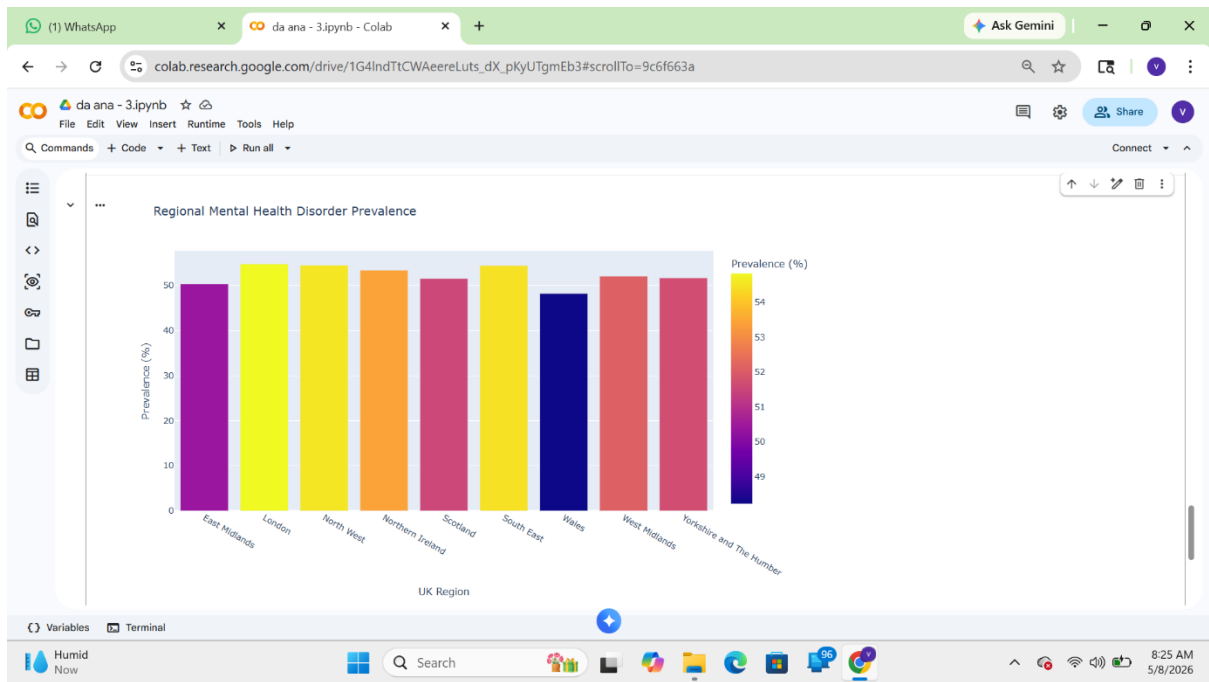
Choropleth Map





Regional Prevalence





Conclusion

The Regional Prevalence Mapping analysis successfully examined the distribution of mental health disorders such as anxiety and depression across different regions of the United Kingdom using the APMS-style dataset. The study demonstrated how geographic analysis can help identify regional variations in mental health prevalence and provide meaningful insights for healthcare planning and policy development.

Using Python, Pandas, Matplotlib, and Google Colab, the dataset was processed, grouped region-wise, and visualized through choropleth mapping techniques.

The choropleth map provided a clear visual understanding of areas with higher and lower prevalence rates.